

DINESH KANNAN

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EDUCATION

North Carolina State University - Master of Computer Science | GPA: 3.85/4.0

August 2023 - May 2025

Coursework: Automated Software Engineering, Cloud Computing, Automated Learning and Data Analysis,

TECHNICAL SKILLS

Programming & Frameworks: Python, JavaScript, React, Node.js, Flask, Java, TypeScript, Rest APIs, Ruby on Rails

Cloud & Databases AWS Cloud, Firebase, Docker, NoSQL, MySQL, Google Cloud, Continuous Integration

Tools: Git, Jira, Heroku, Postman, Power BI, Anaconda, Stripe, Vercel, Linux,

AI & ML : PyTorch, Tensorflow, RAG Pipelines, OpenAI API, Scikit-learn, CNNs, GANs, Transformers

WORK EXPERIENCE

North Carolina State University | Research Assistant

July 2024 - April 2025

- Developed an AI-powered Teaching Assistant chatbot utilizing OpenAI and Retrieval-Augmented Generation (RAG), automating responses and enhancing student engagement, leading to a 35% increase in resource accessibility.
- Implemented a human-in-the-loop learning pipeline integrated with course materials, prompting students for initial problem solving attempts, and providing detailed AI feedback achieving a 40% improvement in student comprehension and critical thinking compared to traditional AI methods.
- Built a responsive full-stack application (React, Flask, MongoDB) featuring JWT authentication, query caching, and embedded academic content, boosting chatbot accuracy by 30%.

Headstarter AI | Software Engineering Fellow

July 2024 - September 2024

- Built core components of a SaaS app using Next.js and OpenAI API, including AI-based QA interface and backend data handling.
- Developed SaaS flashcard app with secure OAuth2-based login and Stripe checkout integration, reducing signup + payment flow from 1m20s to under 1m (22% faster).
- Automated CI/CD workflows via GitHub Actions and Vercel, decreasing deployment time by 25% through streamlined build and testing processes.

Deloitte | Deloitte Consulting Analyst

January 2023 - July 2023

- Software Development-** Engineered five JavaScript web API scripts for Dynamics CRM, improving data retrieval speeds by 40% and enhancing security for multiple enterprise clients.
- Automated workflows using Power BI and JavaScript, achieving a 35% efficiency improvement across teams.
- Contributed to Agile sprints, implementing prioritized features collaboratively and reducing feature delivery timelines by 27%.
- Helped design and implement hybrid MySQL + MongoDB schema to support real-time analytics with reduced query latency.

PROJECTS

Ellipsis — Podcast Generation Platform | Python, Flask, React, Redis, Orpheus TTS

- Developed a fully automated, AI-driven podcast generation agent capable of producing human-like, high-quality podcast on diverse topics and recent news using Perplexity's deep research API.
- Integrated intelligent multi-speaker dialogue generation, significantly enhancing conversational realism and user engagement.

BurnoutExtreme | Flask, React.js, MongoDB, Google Maps API

- Built an integrated fitness platform with modular components for diet tracking (MongoDB), real-time step monitoring, and AI-generated workout plans, increasing user engagement by 27%.
- Implemented an AI chatbot for personalized meal and workout suggestions using user health data, improving recommendation accuracy by 30% along with integrating Google Maps API to enable users to discover and participate in local fitness events, fostering community engagement.
- Improved user retention by 33% by implementing MongoDB-based meal tag filters, chart-based progress visualization (Chart.js), and persistent custom workout plans tied to user profiles.

Sample Efficient Rapid Thermal Modeling of 3D IC | PyTorch, Diffusion Models

- Developed a conditional diffusion model in PyTorch for predicting high-resolution thermal maps from power maps, achieving a 56% performance improvement over traditional GAN models.
- Optimized model architecture to handle intricate thermal distributions and diverse 3D IC stack layouts, including TSVs and routing congestion effectively.

FoodZ Social Media Platform for Food Vloggers – Angular.js, MongoDB, Node.js, AWS Lambda

- Developed a food vloggers social media platform with Angular.js and MongoDB, featuring channels, video uploads, and real-time notifications via WebSockets, slashing message latency by 30%.
- Designed AWS Lambda functions for real-time notifications in a serverless architecture, cutting server costs by 30% and ensuring seamless scalability during user activity spikes.
- Integrated an automated system to analyze 1,000+ user-submitted ingredients, providing users with effortless access to comprehensive nutritional information and enhancing the overall food-related upload experience.